

Practice Questions for 13L1-Logic

Generated by Scribe.AI

December 18, 2024

1 Questions

1. According to the lecture, what is the primary role of logic in automated reasoning?
 - A. To provide a set of rules for learning from data
 - B. To serve as a language for expressing assumptions and conclusions
 - C. To define the architecture of intelligent agents
 - D. To optimize the performance of search algorithms
 - E. To create visual representations of data
2. According to the lecture, which of the following is NOT a logical connective used in propositional logic?
 - A. Conjunction ($\wedge$)
 - B. Disjunction ($\vee$)
 - C. Implication ($\Rightarrow$)
 - D. Biconditional ($\Leftrightarrow$)
 - E. Quantification ($\forall$)
3. According to the lecture, what does it mean for a propositional logic sentence to be "satisfiable"?
 - A. It is true in all possible models.
 - B. It is false in all possible models.
 - C. It is true in at least one model.
 - D. It is equivalent to another sentence.
 - E. It contains no logical connectives.

2 Answers

1. According to the lecture, what is the primary role of logic in automated reasoning?
 - A. Answer A is incorrect because logic is not primarily used for learning from data, but for reasoning.
 - B. Answer B is correct because the lecture states that logic is the language of automated reasoning, used to express assumptions and conclusions.
 - C. Answer C is incorrect because logic is not used to define the architecture of intelligent agents, but to enable reasoning.
 - D. Answer D is incorrect because logic is not used to optimize search algorithms, but to enable reasoning.
 - E. Answer E is incorrect because logic is not used to create visual representations of data, but to enable reasoning.
2. According to the lecture, which of the following is NOT a logical connective used in propositional logic?
 - A. Answer A is incorrect because conjunction is a logical connective used in propositional logic.
 - B. Answer B is incorrect because disjunction is a logical connective used in propositional logic.
 - C. Answer C is incorrect because implication is a logical connective used in propositional logic.
 - D. Answer D is incorrect because biconditional is a logical connective used in propositional logic.
 - E. Answer E is correct because quantification is not a logical connective used in propositional logic, but in predicate logic.
3. According to the lecture, what does it mean for a propositional logic sentence to be "satisfiable"?
 - A. Answer A is incorrect because a sentence that is true in all possible models is considered valid, not just satisfiable.
 - B. Answer B is incorrect because a sentence that is false in all possible models is considered unsatisfiable, not satisfiable.
 - C. Answer C is correct because a sentence is satisfiable if it is true in at least one model.
 - D. Answer D is incorrect because logical equivalence is a different concept from satisfiability.
 - E. Answer E is incorrect because satisfiability is a property of sentences that can contain logical connectives.